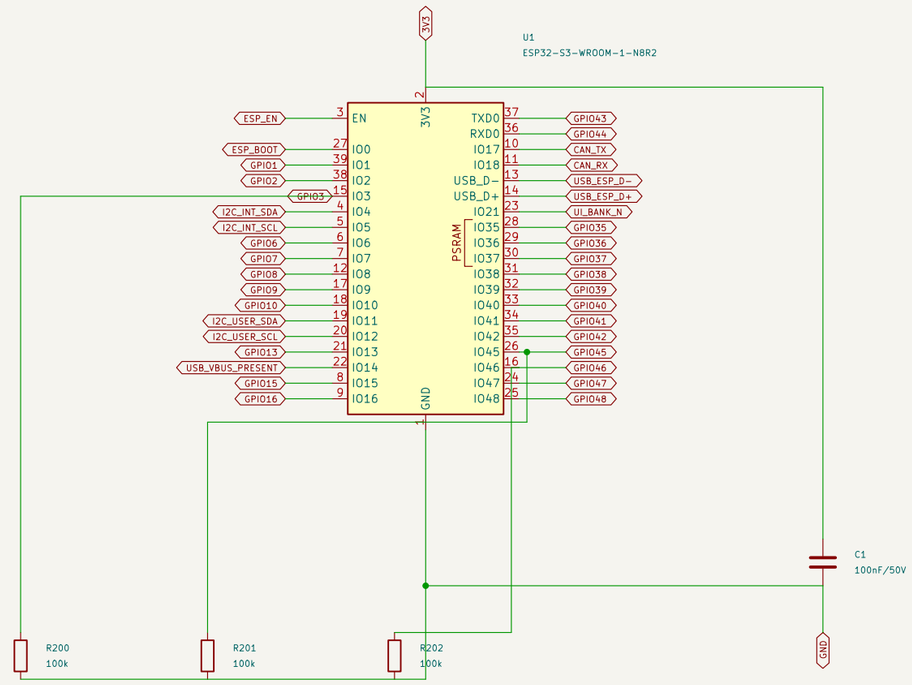
SPAGHETTI LAB | 16 RIGHE | ESP32-S3 N8R2

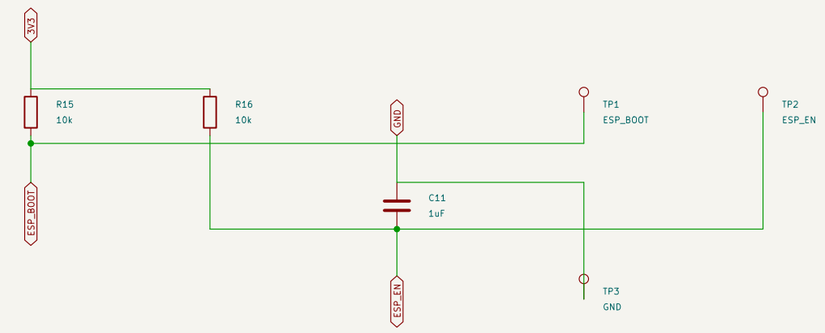
ESP32 / FLEXY PIN

GPIO1..10: ADC1. GPIO35..42: digitali. USB e CAN dedicati.



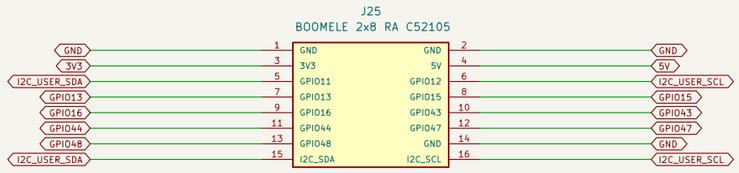
BOOT, EN E PIAZZOLE DI SERVIZIO

ESP_BOOT=strap GPIO0: ESP_EN=reset. Pull-up mantenuti.



ESPANSIONE SUD / BOOMELE C52105 2x8

GPIO diretti 3.3V. SDA/SCL condivisi con Qwiic ed expander.



Pin1 quadrato = GND. Passo e distanza tra file: 2.54mm.

5V solo alimentazione; GPIO non tolleranti 5V. Nessuna commutazione delle righe.

LEGENDA COLLEGAMENTI GLOBALI
3V3: rail regolata; 5V: rail ingresso; GND: riferimento comune.
I2C_INT: GPIO4/5 solo expander; I2C_USER: GPIO11/12 connettori; CAN_TX/RX: transceiver.
USB_ESP_D+/D-: coppia USB verso ESP; USB_VBUS_PRESENT: rilevamentoSVUSB.
RGB_3V3_DATA: uscita ESP per LED; EXP_RESET_N: reset comune expander.
ROW_Lxx = banco A; ROW_Rxx = banco B. Suffissi spiegati nel foglio05.
Tutti i nomi sono globali; i fili mostrano la topologia. Punto = connessione, incrocio senza punto = nessun collegamento.

02 – Alimentazione e USB

File: power-usb.kicad_sch

03 – CAN e terminazione automatica

File: can.kicad_sch

04 – Controllo righe, reset sicuro e Qwiic

File: control.kicad_sch

05 – Sedici righe programmabili

File: ports16.kicad_sch

06 – Indicatori di configurazione retroilluminati

File: rgb-ring.kicad_sch

07 – Mezzanina passiva

File: mezzanine.kicad_sch

08 – Selezione, comandi e logo

File: ui-indicators.kicad_sch

09 – Row touch,16keys

File: touch16.kicad_sch

10 – Global touch commands

File: touch-global.kicad_sch

Global commands: U91, independent software I2C on GPIO21/GPIO13. No physical buttons.
Row touch: U90,16keys on internal I2C. Current PCB is an unrouted design checkpoint.

11 – Blue logo,16 LEDs

File: logo16.kicad_sch

Sheet: /
File: backbone.kicad_sch

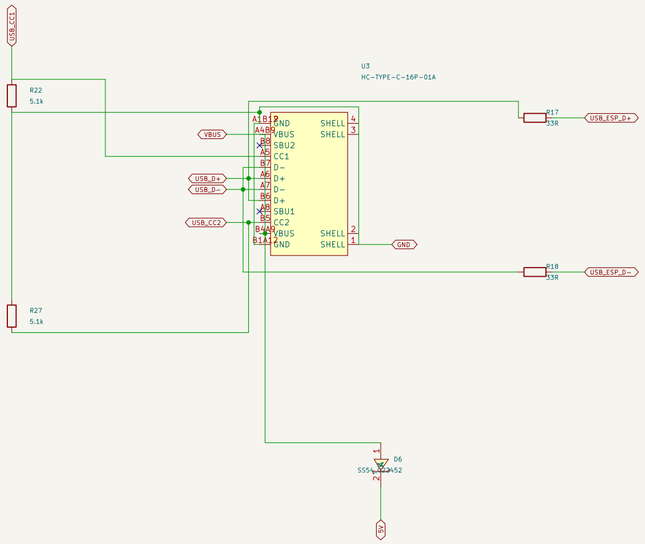
Title: 01 – ESP32 e boot

Size: User | Date: 2026-11-09 | **Rev: 16R-SELECTOK4**
KiCad E.D.A. 10.0.3 | Id: 1/11

USB-C E ALIMENTAZIONE 5V / 3V3

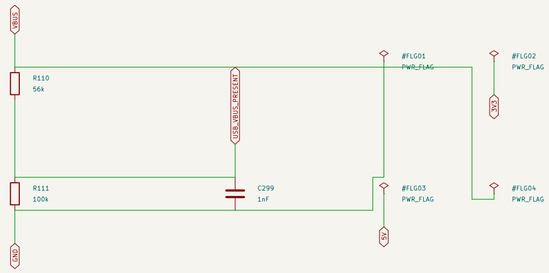
USB-C, CC E DATI

CC1/CC2: resistenze dispositivo USB; D6: alimentazione da VBUS.



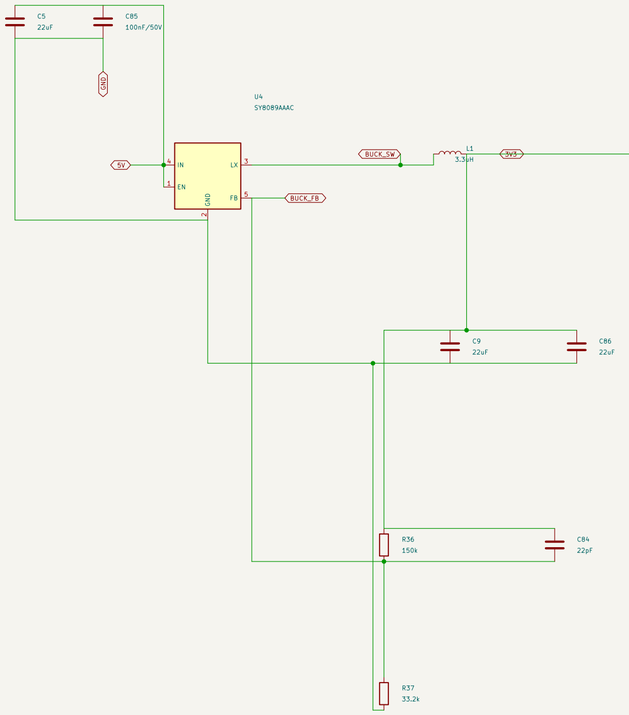
RILEVAMENTO VBUS E FLAG ALIMENTAZIONI

USB_VBUS_PRESENT: partitore56k/100k verso GPIO14, filtro1nF.



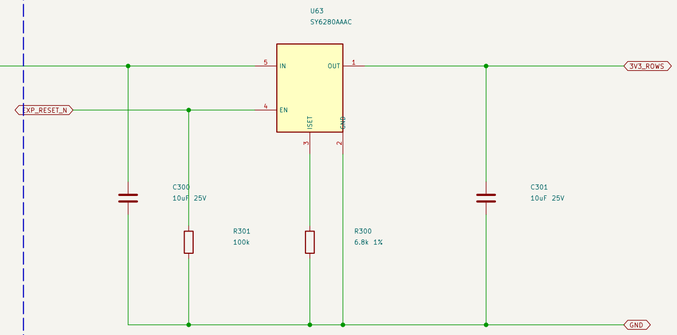
REGOLATORE 5V -> 3V3

BUCK_SW: nodo commutato; BUCK_FB: retroazione del regolatore.
Le etichette 5V/3V3/GND connettono gli altri fogli.



PROTEZIONE COMUNE ALIMENTAZIONE RIGHE

3V3_ROWS alimenta solo i rami VCC delle16righe.
ESP32, CAN, Qwiic e connettori diretti restano su3V3.



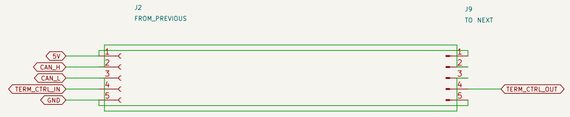
R300: ILIM=6800/R; 6.8k -> 1A nominale condiviso.
Datasheet:0.75..1.25A a5V/25C;500mA e3.3V da validare.
Corto persistente: limite corrente, arresto termico e riprova.
Non protegge GPIO,5V esterni o le rail dirette dei connettori.
Il chip non fornisce un segnale FAULT. EN non deve fluttare.

Avvio: EXP_RESET_N basso -> U63 spento.
Senza scarica attiva (versioneA): I condensatori possono restare carichi. Righe Hi-Z garantite solo dalle sequenze e dai pull del circuiti esistenti, da provare sul prototipo.
GND rimane comune. PCB ancora da sbrogliare.

BUS CAN E TERMINAZIONE AUTOMATICA

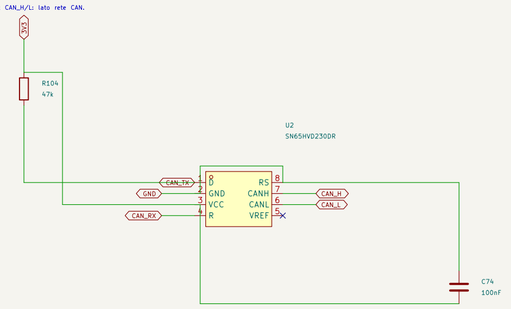
CONNETTORI MAGNETICI PASSANTI

CAN_H/L: bus differenziale. TERM_CTRL_IN/OUT: rilevamento dei vicini.



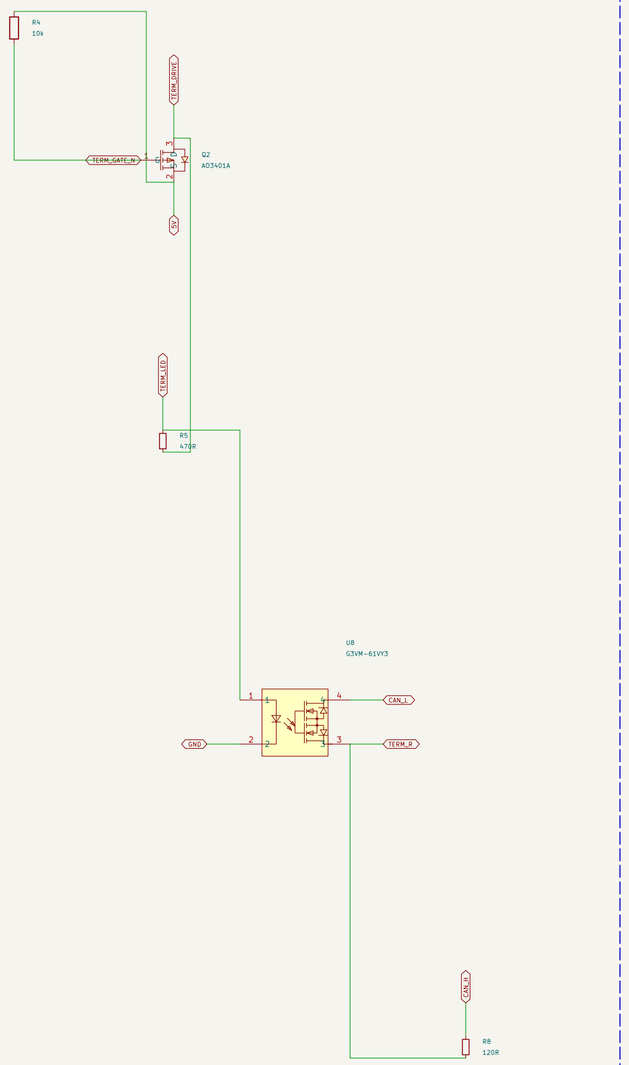
TRANSCEIVER CAN 3V3

CAN_TX/RX: interfaccia ESP: CAN_H/L: lato rete CAN.



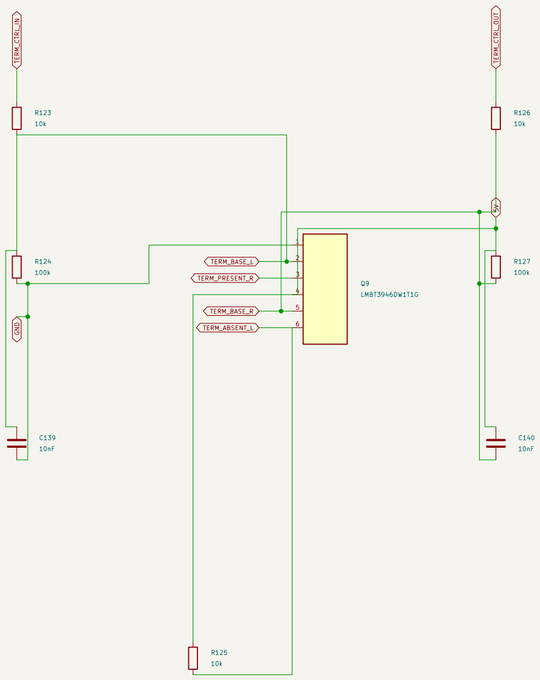
INTERRUTTORE E RESISTENZA DI TERMINAZIONE

TERM_DRIVE / TERM_LED: pilotaggio foto-MOS.
TERM_R: collegamento tra120ohm e interruttore verso CAN_L.
Quando abilitata, la resistenza120ohm e fra CAN_H e CAN_L.



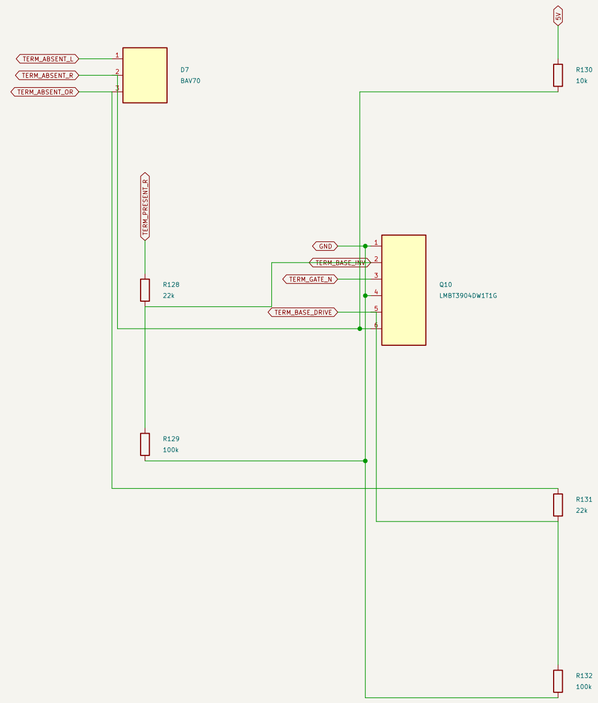
RILEVAMENTO VICINO SINISTRO / DESTRO

TERM_BASE_L/R: nodi di filtro.
TERM_ABSENT_L: assenza a sinistra; TERM_PRESENT_R: presenza a destra.

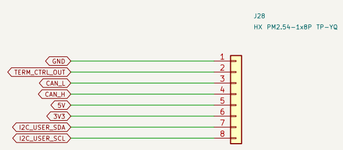


COMBINAZIONE ASSENZE E COMANDO

TERM_ABSENT_R e TERM_ABSENT_OR: estremità rilevate.
TERM_GATE_N: comando attivo basso del PMOS di terminazione.

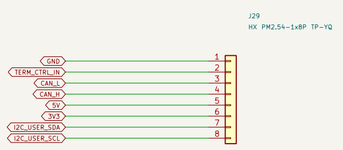


ESPANSIONE BUS J28 / MAGNETICO J9



Pin 1–5 condividono i fori e le reti del magnetico.
6:3V3 / 7:SDA / 8:SCL. I2C solo locale a3.3V.
DET mantiene la funzione originale di terminazione CAN:
non usarlo come GPIO e non forzarlo sulla espansione.
Non alimentare contemporaneamente il bus da sorgenti
non isolate. Nessuna protezione aggiunta su questi pin.

ESPANSIONE BUS J29 / MAGNETICO J2



Pin 1–5 condividono i fori e le reti del magnetico.
6:3V3 / 7:SDA / 8:SCL. I2C solo locale a3.3V.
DET mantiene la funzione originale di terminazione CAN:
non usarlo come GPIO e non forzarlo sulla espansione.
Non alimentare contemporaneamente il bus da sorgenti
non isolate. Nessuna protezione aggiunta su questi pin.

Sheet: /03 – CAN e terminazione automatica/
File: can.kicad_sch

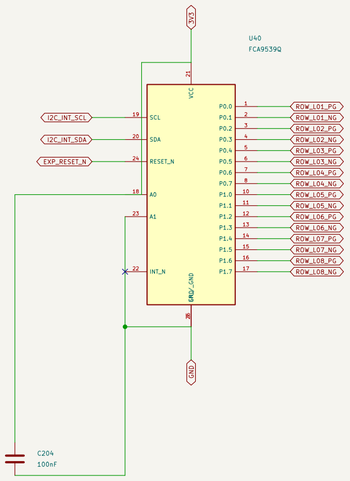
Title: 03 – CAN e terminazione automatica

Size: User	Date: 2026-11-07	**Rev: 16R-READABLE**
KiCad E.D.A. 10.0.3		Id: 3/11

CONTROLLO48SEGNALI / RESET SICURO / I2C

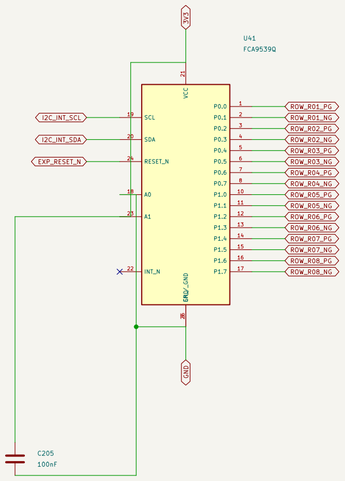
U40 | FCA9539Q | 0x75

PG/NG banco A: comandi di alimentazione.
SCL/SDA: I2C; A0/A1: indirizzo; INT_N non usato.



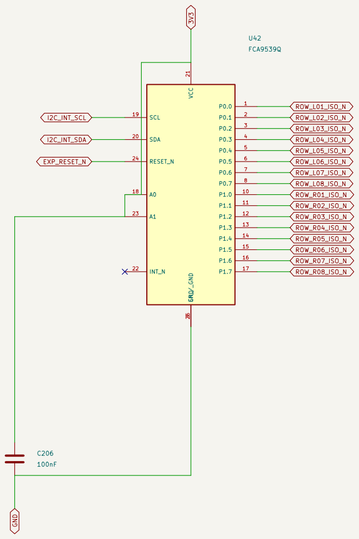
U41 | FCA9539Q | 0x76

PG/NG banco B: comandi di alimentazione.
SCL/SDA: I2C; A0/A1: indirizzo; INT_N non usato.



U42 | FCA9539Q | 0x77

ISO_N A/B: isolamento GPIO, attivo basso.
SCL/SDA: I2C; A0/A1: indirizzo; INT_N non usato.



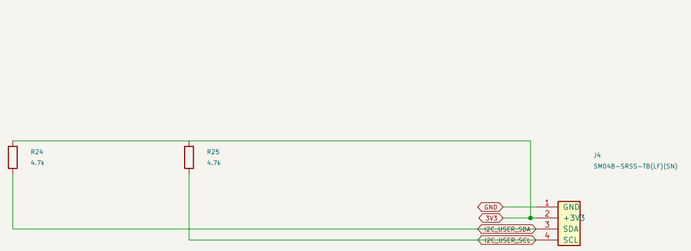
INTERBLOCCO RESET / UNA SOLA PORTA

EXP_RESET_N = ESP_EN AND NOT GPIO0. INO aGND.
R203 mantiene reset basso se non pilotato; C200 bypass100nF.

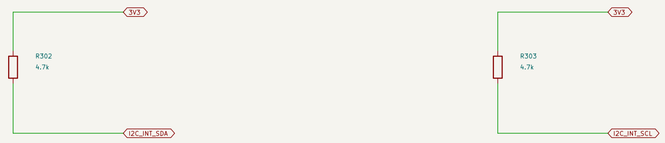


BUS UTENTE GPIO11/12 / QWIIC / PULL-UP

I2C_USER indipendente dall'expander: SDA GPIO11 / SCL GPIO12. Pull-up4.7k a3V3.
3V3/GND: alimentazione del moduloQwiic.



BUS INTERNO DEDICATO / GPIO4 SDA / GPIO5 SCL



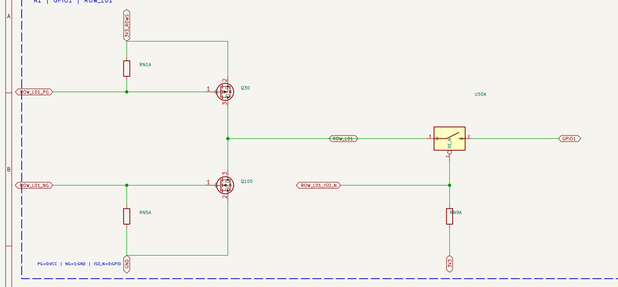
R302/R303: pull-up4.7k del solo bus interno.
R24/R25: pull-up4.7k del bus utente, nel blocco Qwiic.
Nessun collegamento SDA/SCL tra i due bus.
Entrambi usano livelli3.3V: firmware: due controller distinti.

INTERFACCIA LOCALE | DUE PULSANTI XKB

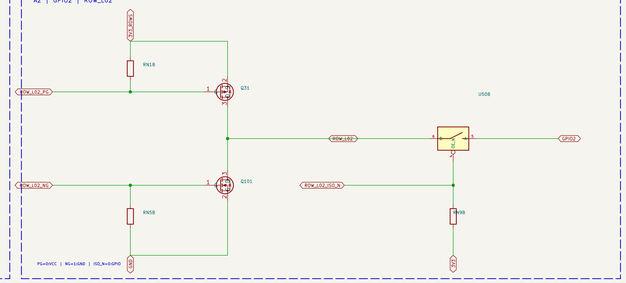
Sheet: /04 – Controllo righe, reset sicuro e Qwiic/	
File: control.kicad_sch	
Title: 04 – Controllo righe, reset sicuro e Qwiic	
Size: User	Date: 2026-11-08
KiCad E.D.A. 10.0.3	**Rev: 16R-READABLE**
	Id: 4/11

16 RIGHE PROGRAMMABILI | VCC3V3 / GND / GPIO / HI-Z

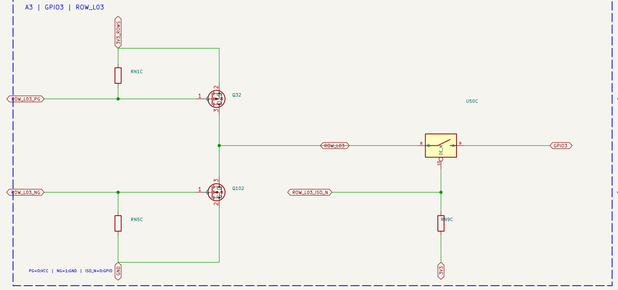
A1 | GPIO1 | ROW_L01



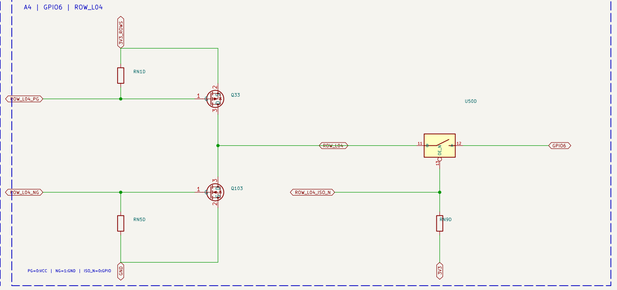
A2 | GPIO2 | ROW_L02



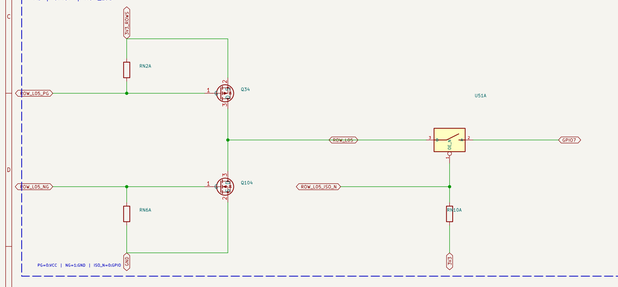
A3 | GPIO3 | ROW_L03



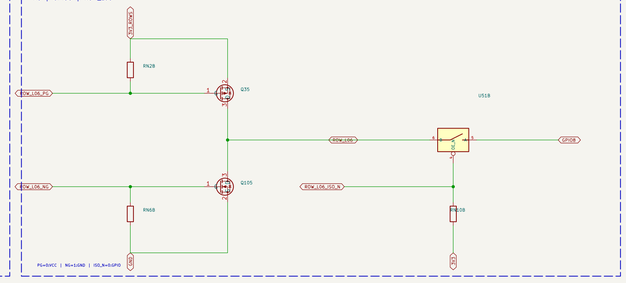
A4 | GPIO6 | ROW_L04



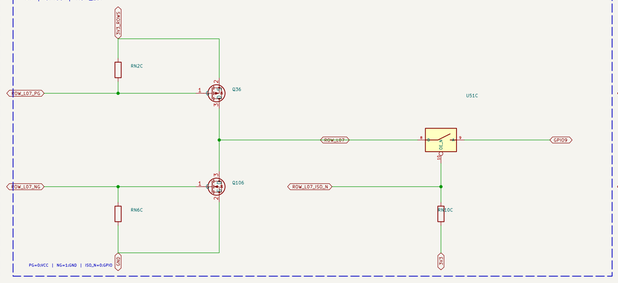
A5 | GPIO7 | ROW_L05



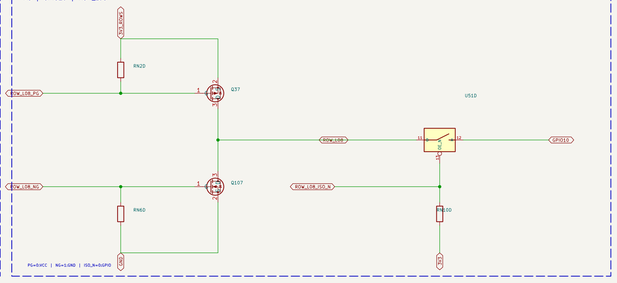
A6 | GPIO8 | ROW_L06



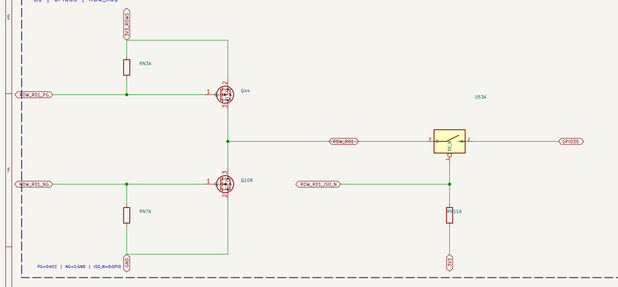
A7 | GPIO9 | ROW_L07



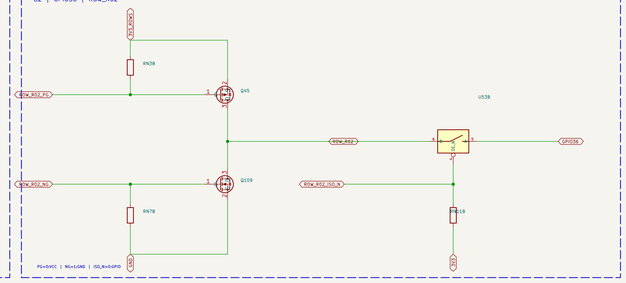
A8 | GPIO10 | ROW_L08



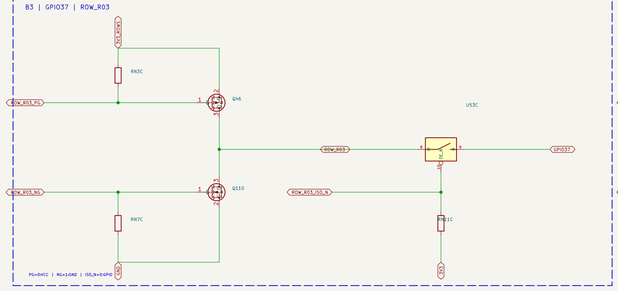
B1 | GPIO35 | ROW_R01



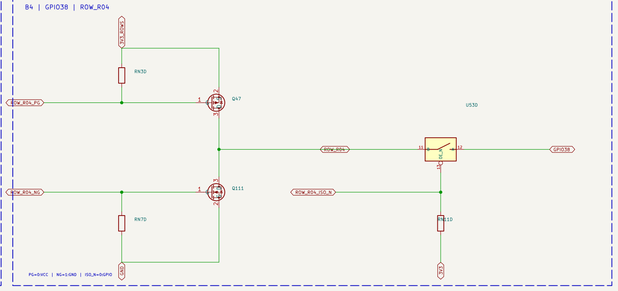
B2 | GPIO36 | ROW_R02



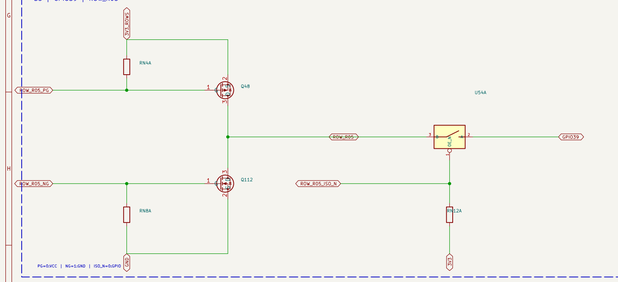
B3 | GPIO37 | ROW_R03



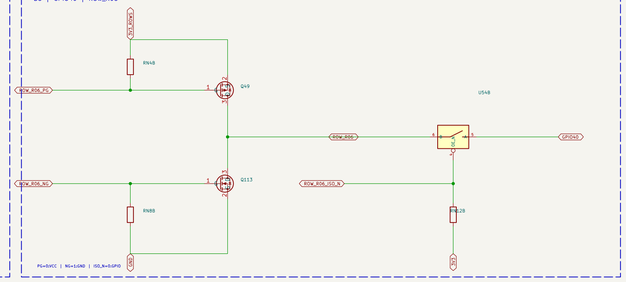
B4 | GPIO38 | ROW_R04



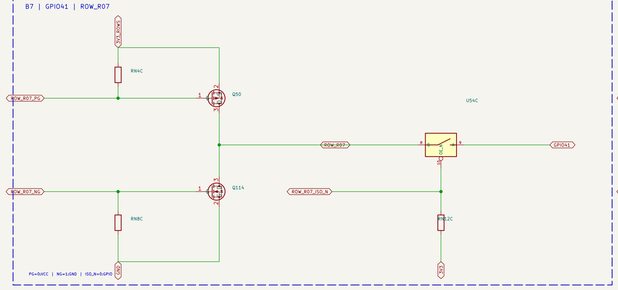
B5 | GPIO39 | ROW_R05



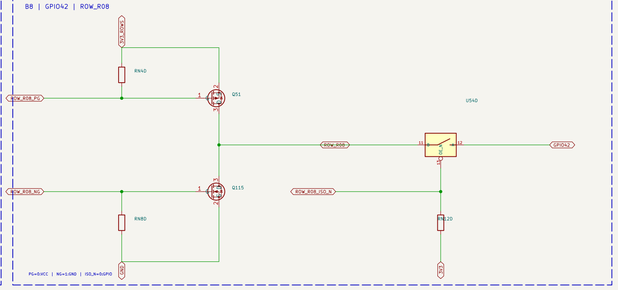
B6 | GPIO40 | ROW_R06



B7 | GPIO41 | ROW_R07

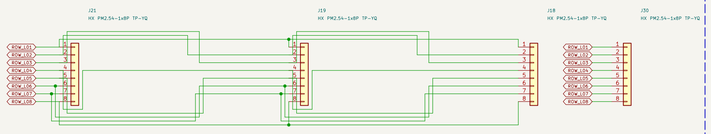


B8 | GPIO42 | ROW_R08



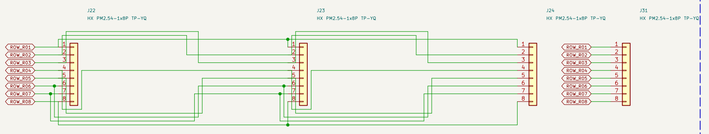
SOCKET BANCO A | 4CONTATTI COMUNI PER RIGA

ROW_L01..08=A1..A8. ROW_R01..08=B1..B8. Passo righe2.54mm; colonne4mm; separazione centrale7.62mm.



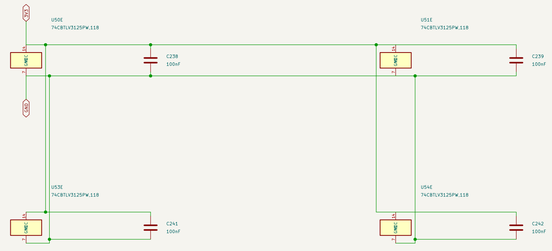
SOCKET BANCO B | 4CONTATTI COMUNI PER RIGA

ROW_L01..08=A1..A8. ROW_R01..08=B1..B8. Passo righe2.54mm; colonne4mm; separazione centrale7.62mm.



ALIMENTAZIONE DEI4ISOLATORI

Ogni integrato bancasali. UnitaE=alimentazione: bypass100nF.



REGOLA: PG=1, NG=0, ISO_N=1 -> Hi-Z. Per ADC: disattivare entrambi i MOSFET, utilizzare solo ISO_N e ingresso analogico.
Massa comune: tutti i GND collegati insieme. Alimentazione: +3.3V_ROW e GND. Connettore: Molex Micro-Fit 3.0 43650-xx16 (M) e 43640-xx16 (F). Switch bidirezionali TS5DP. Inserire 100nF vicino a ogni IC.
Q30..Q51 = AO3416 (P); Q100..Q115 = AO3400A (N). Switch bidirezionali TS5DP. Inserire 100nF vicino a ogni IC.
Obiettivo 500 mA su singola riga in VCC/GND: budget condiviso, piste da sbroglliare e prova termica necessaria. GPIO/ADC non sono uscite da 500 mA.

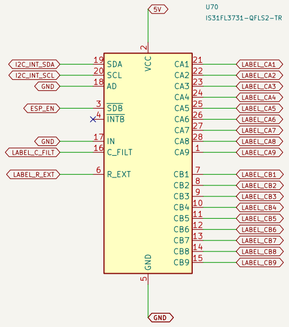
Sheet: /05 – Sedici righe programmabili/
File: porte16.kicad_sch

Title: 05 – Sedici righe programmabili

Size: A1	Date: 2026-11-07	Rev: 16R-READABLE
KiCad E.D.A. 10.0.3		Id: 5/11

DRIVER5V | I2C INTERNO3,3V |48LED

is31fl3731_0x74: SDB=ESP_EN3,3V. Nessun nuovo GPIO. R_EXT47kΩ.38mAmmedi/LED aPWM255.
CA16rossi, CB:16blu+16verdi. Avvio in software shutdown; azzerare enable/PWMprima di accendere.











A1 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

A2 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

A3 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

A4 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

A5 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

A6 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

A7 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

A8 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

B1 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

B2 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

B3 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

B4 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

B5 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

B6 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

B7 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

B8 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

Sheet: /06 – Indicatori di configurazione retroilluminati/
File: rgb-ring.kicad_sch

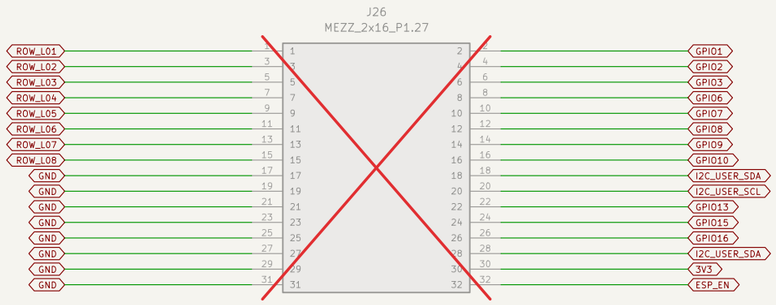
Title: 06 – Indicatori tre colori,5V,testi dritti

Size: A2	Date: 2026-11-08	**Rev: 16R-READABLE**
KiCad E.D.A. 10.0.3		Id: 6/11

J26 / BANCO A: FORI E GPIO SEPARATI

1..16: ROW = fori; GPIO = lato ESP, prima dello switch.

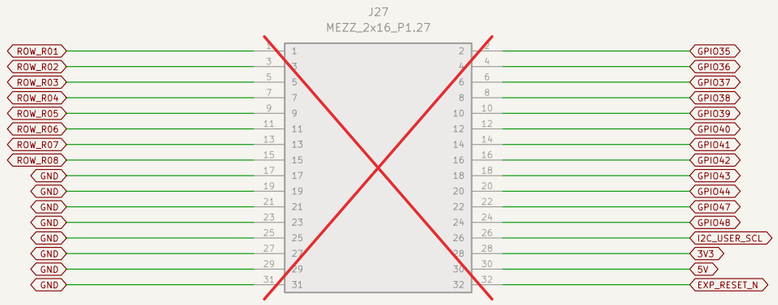
17..32: espansione, rail, masse e reset.



J27 / BANCO B: FORI E GPIO SEPARATI

1..16: ROW = fori; GPIO = lato ESP, prima dello switch.

17..32: espansione, rail, masse e reset.



BACKBONE PASSIVA: aggiunti solo J26/J27: nessun ADC/DAC, selettore, shunt o protezione extra.
Mezzanina assente: prese aperte. Presente: ingressi/uscite analogiche da progettare ad alta impedenza e sicuri anche senza alimentazione.
GPIO e ROW non vanno uniti sulla mezzanina: bypasserebbero gli switch e la separazione dal3V3/GND delle righe.
GPIO3.3V, non tolleranti5V. Righe limitate al dominio della backbone. I2C_USER condiviso con Qwiic/Hanxia, separato dagli expander; evitare conflitti di indirizzo.
ESP_EN: reset ESP, non abilitazione dedicata. EXP_RESET_N: uscitaU60, solo lettura; non comandarla dalla mezzanina.
Sincronizzazione e abilitazione: assegnabili nel firmware ai7GPIO liberi esposti anche suJ25; GPIO11/12 riservati al bus utente, non a pin aggiuntivi dedicati.
La mezzanina di misura deve avere protezioni, disconnessione a riposo e sequenze sicure proprie; nessuna protezione nuova sulla base.
La corrente si misura instradando il circuito fra due righe distinte attraverso lo shunt della mezzanina; nessuno shunt sulla backbone.

Sheet: /07 – Mezzanina passiva/
File: mezzanine.kicad_sch

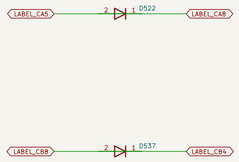
Title: 07 – Mezzanina passiva / tutti i nodi e GPIO

Size: A3	Date: 2026-11-08	**Rev: MEZZ-PASSIVE**
KiCad E.D.A. 10.0.3		Id: 7/11

BLUE LOGO: see sheet11. Sixteen emitters, existing driver: no new GPIO or LED driver.

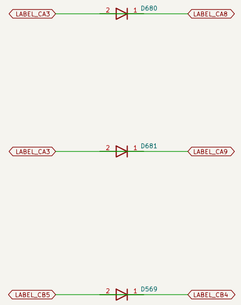
PLAY/PAUSE AND DEBUG

Driven by existing IS31FL3731; global touch inputs are on sheet10.



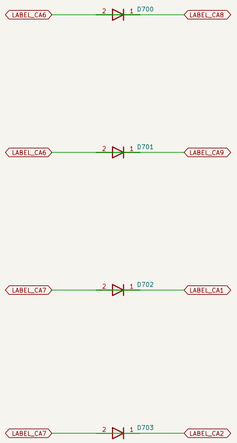
LOAD / SAVE / DELETE

Driven by existing IS31FL3731; global touch inputs are on sheet10.



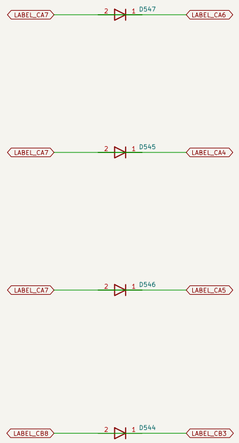
CONFIGURATION SLOTS

Driven by existing IS31FL3731; global touch inputs are on sheet10.



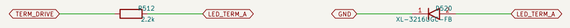
CONNECTION STATUS

Driven by existing IS31FL3731; global touch inputs are on sheet10.



CAN TERMINATION

Existing direct-drive LED retained.



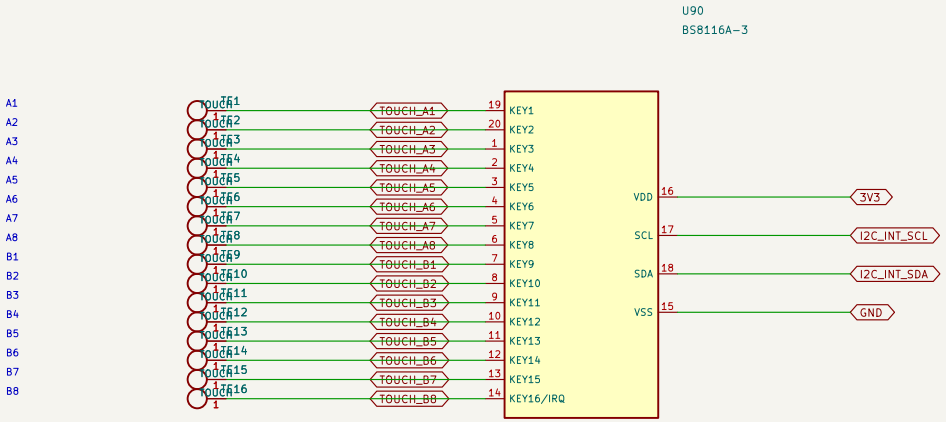
Sheet: /08 – Selezione, comandi e logo/
File: ui-indicators.kicad_sch

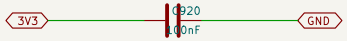
Title: 08 – LED status and touch command indicators

Size: A2	Date: 2026-11-07	**Rev: 16R-READABLE**
KiCad E.D.A. 10.0.3		Id: 8/11

ONE CONTROLLER / SIXTEEN ROW KEYS

U90 BS8116A-3, address0x50 on internal I2C. No multiplexer.
Key16 defaults to IRQ: clear K16_TH bit6 (0xC4), use polling. Follow full configuration-write protocol.



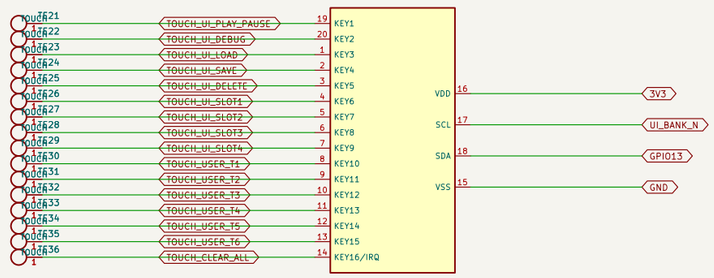


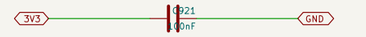
Global actions use U91 on dedicated GPIO21/GPIO13 software I2C.
Touch discrimination, full routing and final hardware qualification remain open.

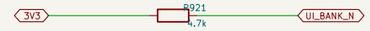
Sheet: /09 - Row touch,16keys/		
File: touch16.kicad_sch		
Title: 09 - Sixteen row touch keys		
Size: A4	Date: 2026-11-09	**Rev: 16R-SELECTOK4**
KiCad E.D.A. 10.0.3		Id: 9/11

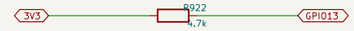
TEN COMMANDS + SIX USER TOUCH KEYS / BS8116A-3

U91 fixed address0x50 on GPIO21 SCL / GPIO13 SDA (software I2C). U90 stays on internal I2C.
External4.7k pull-ups to3.3V. All16 keys used. Key16 defaults to IRQ: clear K16_TH bit6 (0xC4). follow full configuration write protocol.









PLAY/PAUSE: tap toggles; hold2s requests STOP/Hi-Z. DEBUG: request analysis mode.
LOAD/SAVE: select action, then slot1..4. DELETE: hold slot2s. No physical switches.
Touch discrimination, full copper routing and firmware integration remain to be validated.

CLEAR ALL: hold2seconds. Disconnect every row (Hi-Z), clear working configuration, cancel execution. Saved slots are retained. KEY16/IRQ must be configured as a touch key.

Physical layout: user T1..T6 in an aligned3x2 grid at upper left. LOAD / SAVE / DELETE / CLEAR at bottom right, with slots1..4 aligned below.

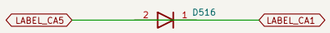
Sheet: /10 – Global touch commands/
File: touch-global.kicad_sch

Title: 10 – Global and programmable user touch controls

Size: A3	Date: 2026-11-07	**Rev: 16R-READABLE**
KiCad E.D.A. 10.0.3		Id: 10/11

SPAGHETTI LAB /16 BLUE REVERSE-MOUNT LEDS

D516..D519 retained: D720..D731 added. All16 rotated45degrees,3mm pitch.
Directed driver pairs are unique:77/144 channels populated,67free. No additional driver.

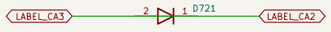






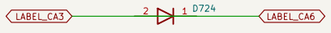


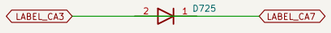


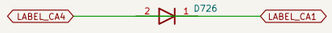






















Logo geometry traced from the user-supplied white-background original with preserved aspect ratio.
Light distribution through FR4, emitter-side routing, final LED current and optical uniformity require prototype qualification.

Sheet: /11 – Blue logo,16 LEDs/	
File: logo16.kicad_sch	
Title: 11 – Blue logo,16 emitters on existing LED driver	
Size: A3	Date: 2026-11-07
KiCad E.D.A. 10.0.3	**Rev: 16R-READABLE**
	Id: 11/11